FINC5001 FOUNDATION IN FINANCE

TOPIC 3 – FINANCIAL MATHEMATICS II – QUESTIONS AND SOLUTIONS

PRE-WORKSHOP QUESTIONS

Pre-workshop self-access questions set is designed for you to self-access your understanding of the lecture content. You are expected to complete these questions prior to the workshop. These questions will not be discussed in the workshop.

Pre-Q1

Suppose you deposit $500 in account at the end of each month at 12% compounded monthly. How much will you have in your account in 5 years' time?

This question is asking for the future value of an ordinary annuity
First determine the effective rate per month:

$$Effective\ Rate = \left(1 + \frac{0.12}{12}\right)^{12x\frac{1}{12}} - 1 = 0.01$$

Then use the future value of an annuity to determine the future value in 5 years' (60 monthly periods) time

$$FV = PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

$$FV_5 = 500\left[\frac{1.01^{60} - 1}{0.01}\right] = \$40{,}834.83$$

Pre-Q2

You are considering a potential investment that will generate a cash flow of $1000 every six months for 5 years with the first cash flow to be received immediately (a total of ten cash flows). If the appropriate discount rate is 12% pa compounded quarterly, what is the intrinsic value of the investment?

This question is asking for the present value of the annuity due
First determine the effective rate per half year:

$$Effective\ Rate = \left(1 + \frac{0.12}{4}\right)^{4x\frac{1}{2}} - 1 = 0.0609$$

Then use the present value of an annuity due to determine the present value of the cash flow stream

$$PV = PMT + PMT\left[\frac{1-(1+i)^{-(n-1)}}{i}\right]$$

$$PV_0 = 1000 + 1000\left[\frac{1-(1.0609)^{-9}}{0.0609}\right] = \$7{,}775.13$$

Pre-Q3

You are considering an investment that generates a cash flow of $1000 next year. The investment is then expected to generate annual cash flows that grow at 5% pa compounded annually for 5 years. If the required rate of return is 10% pa compounded annually, what is the intrinsic value of this investment opportunity?

This is a growing annuity and can be solved using the growing annuity equation. The solution is given below:

$$PV = PMT \times \frac{1}{i-g}\left[1 - \frac{(1+g)^n}{(1+i)^n}\right]$$

$$PV = 1000 \times \frac{1}{0.1-0.05}\left[1 - \frac{(1+0.05)^5}{(1+0.10)^5}\right] = \$4150.59$$

Pre-Q4

Your company is deciding between two investment choices: one that pays $100 per year in perpetuity, and another that pays $100 per year for 100 years. The current market interest rate for investments of similar risk is at 10% p.a. What is the present value of these two investments? Why are they so similar?

PV of the perpetuity: $PV = \frac{A}{i} = \frac{100}{0.1} = \$1{,}000$

PV of the annuity: $PV = PMT\left[\frac{1-(1+i)^{-n}}{i}\right] = 100\left[\frac{1-1.10^{-100}}{0.10}\right] = \$999.93$

Given that there are an infinite number of payments for the first investment, its value is not much greater than that of the second investment. This is because cash flows in the far future are discounted very heavily (because of the time value of money) and add very little to the value of the investment.